

Representation theorem for polyhedra

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We study a non-empty polyhedron viewed as a finitely generated set:

$$P = \text{conv}(V) + \text{cone}(E),$$

where $V, E \subset \mathbb{R}^n$ are finite non-empty sets. The set $\text{rec}(P) = \text{cone}(E)$ is called the *recession cone* of P . A convex cone is called *pointed* iff it does not contain a line through the origin (a 1-dimensional linear space).

Extreme directions are directions in a convex cone that are not conic combinations of other (non-equivalent) directions in the cone. For a pointed polyhedral cone, the extreme directions are the essential generators (proved in note 1.(G)):

Fact 1 (*Representation theorem for pointed polyhedral cones*)

Let $C = \text{cone}(S)$ be a pointed polyhedral cone, where $S \subset \mathbb{R}^n$ is finite and non-empty. If $C \neq \{0_n\}$, there exists a non-empty set $S^ \subseteq S$ such that $C = \text{cone}(S^*)$ and all directions in S^* are extreme directions for C . ■*

An extreme point of P is a point $x \in P$ that is not a convex combination of points in $P \setminus \{x\}$. In this note we show that, if $\text{rec}(P)$ is pointed, then V can be chosen to be the extreme points of P , and E can be chosen to be extreme directions of $\text{rec}(P)$. This generalizes Fact 1.

The key idea is to apply Fact 1 to the cone extension of P :

$$C_f(V, E) := \text{cone}\left(\left\{\begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V\right\}\right) + \text{cone}\left(\left\{\begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E\right\}\right).$$

The polyhedron P is connected to its cone extension $C_f(V, E)$ by the one-to-one relationship $\begin{pmatrix} x \\ 1 \end{pmatrix} \in C_f(V, E) \iff x \in P$.

It is only possible to apply Fact 1 to $C_f(V, E)$ when $C_f(V, E)$ is pointed. One has that $C_f(V, E)$ is pointed exactly when $\text{rec}(P)$ is pointed (Theorem 1 in note 2.(D)).

Fact 2 (*Pointedness of $\text{rec}(P)$ implies pointedness $C_f(V, E)$*)

The cone extension $C_f(V, E)$ is pointed $\iff$ the cone $\text{rec}(P)$ is pointed. ■

In the remainder we assume $\text{rec}(P)$ is pointed. Fact 1 then shows there exists finite sets $V^*, E^* \subset \mathbb{R}^n$ not both empty such that:

$$C_f(V, E) = C_f(V^*, E^*) = \text{cone}(\{ \begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V^* \}) + \text{cone}(\{ \begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E^* \}),$$

and such that the directions in $\{ \begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V^* \}$ and $\{ \begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E^* \}$ are extreme directions of $C_f(V^*, E^*)$. (Here we use that extreme directions of $C_f(V, E)$ with positive last coordinate can be scaled to have 1 as last coordinate). We have $P = \text{conv}(V^*) + \text{cone}(E^*)$ with $\text{rec}(P) = \text{cone}(E^*)$.

Observe that V^* is non-empty. Indeed, if one had $V^* = \emptyset$, then $C_f(V, E) = \text{cone}(\{ \begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E^* \})$, which implies that every direction in $C_f(V, E)$ has zero as last coordinate. However, since $\begin{pmatrix} x \\ 1 \end{pmatrix} \in C_f(V, E)$ for every $x \in P$, this contradicts the assumption that P is non-empty.

One has that a point $x \in P$ is an extreme point of P if and only if $\begin{pmatrix} x \\ 1 \end{pmatrix}$ is an extreme direction of $C_f(V, E)$ (Theorem 2 in note 2.(D)):

Fact 3 (*The set V^* consists of the extreme points of P*)

$$v \in P \text{ is extreme for } P \iff \begin{pmatrix} v \\ 1 \end{pmatrix} \text{ is extreme for } C_f(V, E) \iff v \in V^*. \quad \blacksquare$$

Fact 2 and Fact 3 give the representation theorem for polyhedra: If P is non-empty and $\text{rec}(P)$ is pointed, then P is the sum of the convex hull of the extreme points of P , and the conic hull of extreme directions of $\text{rec}(P)$. This gives an essentially unique representation of P .

Theorem 1 (*Representation theorem for polyhedra*)

For any non-empty polyhedron $P \subseteq \mathbb{R}^n$ with $\text{rec}(P)$ pointed, there exists a non-empty and finite set $V^ \subset \mathbb{R}^n$ such that:*

$$(i) \quad P = \text{conv}(V^*) + \text{rec}(P).$$

$$(ii) \quad V^* \text{ is the unique set of extreme points of } P. \quad \blacksquare$$

Since $\text{rec}(P)$ is assumed to be pointed in Theorem 1, Fact 1 shows $\text{rec}(P)$ can be finitely generated by the extreme directions of $\text{rec}(P)$.

We finally observe that Fact 2 and Theorem 1 characterize exactly which polyhedra have extreme points:

Corollary 1 (*Existence of extreme points in polyhedra*)

A non-empty polyhedron P has an extreme point $\iff \text{rec}(P)$ is pointed. $\blacksquare$

Proof: $\Leftarrow$: If P is non-empty and $\text{rec}(P)$ is pointed, then Theorem 1 shows P has extreme points.

$\Rightarrow$: If $P \neq \emptyset$ and $\text{rec}(P)$ is *not* pointed, there exists $l \in \text{rec}(P) \setminus \{0_n\}$ satisfying $-l, l \in \text{rec}(P)$. For every $x \in P$ one has $x + l, x - l \in P \setminus \{x\}$ and $x = \frac{1}{2} \cdot (x + l) + \frac{1}{2} \cdot (x - l)$, which implies no point in P is extreme for P . $\blacksquare$